


# Reign Performance & Win-Prediction Analysis

Quantitative Analysis of Player Efficiency, Volatility and Team Winning Benchmarks

**Finlay Steel**

Dunfermline Reign Senior Men - 25/26 Season Analytics

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# 1 Executive Summary

This report presents a quantitative evaluation of individual player efficiency, game-to-game performance volatility, and team-level win predictors for the Dunfermline Reign Senior Men's team throughout the 2025/26 basketball season. By breaking down raw multi-sheet game logs, I built advanced performance metrics such as True Shooting Percentage (TS%) and 3-game rolling averages ($MA_3$) to track individual trajectories and form fluctuations across the roster. Combining these individual player metrics into a team-level decision tree model shows that wins depend heavily on hitting specific performance benchmarks. The model highlights three primary baseline targets: achieving a team rebounding total of $> 43.5$ points towards a win for the team. Alternatively, if the team does not rebound well, then the top three scorers in the game combining for $> 57.5$ points and a team TS% higher than 54.1% should still result in a victory for the team.

Additionally, predictive modelling and threshold analysis reveal that team success is heavily dictated by core offensive volume and possession control. Win probability scales dramatically when the team's top three scorers combine to meet required baseline targets, with total rebounding and True Shooting percentages acting as critical safety nets during inefficient shooting nights. These findings provide the coaching staff with clear, evidence-based targets for game strategy, offensive prioritization, and rotation management.

# 2 Dataset & Methodology

## 2.1 Data Ingestion & Cleaning

The dataset covers 25 game logs from the 2025/26 season, taken from `SM Stats 2025-26.xlsx`: 23 competitive fixtures and 2 friendlies. I cleaned and standardised the sheets and removed team-total rows from the player-level analysis. A player counts as having played only if they were on court for at least 7 minutes, and players with fewer than 5 games were left out of the player comparisons.

The friendlies are handled differently depending on the question. They stay in the player-level analysis (Sections 3 and 4), because the minutes and shots are still real and say something about individual form. They are excluded from every team-level measure tied to winning and losing (Sections 2.3, 5 and 6), because friendly results aren't comparable with competitive games. All win rates, box plots and the decision tree therefore use the 23 competitive games only.

## 2.2 Advanced Metric Formulation

To evaluate performance beyond standard box-score totals, several advanced efficiency metrics were derived:

- **True Shooting Percentage (TS%):** Measures overall shooting efficiency by accounting for the added value of three-pointers and free throws:

$$\text{TS\%} = \frac{\text{PTS}}{2 \times (\text{FGA} + 0.44 \times \text{FTA})} \tag{1}$$

- **Assist-to-Turnover Ratio (AST/TOV):** Measures decision-making and ball security while preventing division-by-zero errors:

$$\text{AST/TOV} = \frac{\text{AST}}{\max(1, \text{TOV})} \tag{2}$$

- **3-Game Rolling Moving Average ($MA_3$):** Smooths single-game variance to track short-term form and efficiency trajectories across consecutive fixtures:

$$\mathrm{MA}_3(t) = \frac{1}{3}\sum_{i=0}^{2} x_{t-i} \tag{3}$$

where $t$ denotes the index of the current game and $x_t$ represents the metric value achieved in that fixture. Note that the first two games use a shorter window, so the early values are one and two-game figures.

### 2.3 Exploratory Data Analysis & Team Baselines

To establish a clear baseline before looking at individual players, team averages were split and compared across wins and losses. Comparing these two outcomes highlights the most significant statistical differences driving game results, specifically around scoring efficiency, rebounding, and ball care.

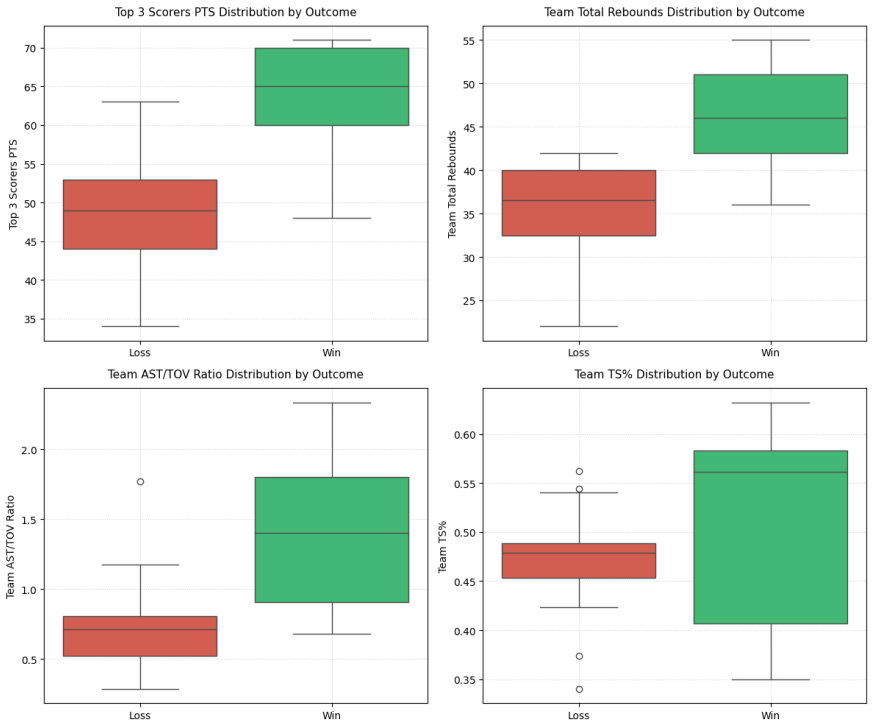



Figure 1: Team Statistical Comparison: Performance Metrics in Wins vs. Losses.

As shown in Figure 1, comparing winning games to losing games reveals clear performance gaps. Winning performances are defined by increases in overall shooting efficiency (TS%), rebounding and better assist-to-turnover ratios (AST/TOV). The most noticeable increase being the combined points of the top 3 scorers. Establishing these team-level baselines gives important context before evaluating individual player profiles and win probability models in later sections.

# 3 Player Profiling: Volume vs. Efficiency

## 3.1 Bivariate Distribution & Baseline Mechanics

Evaluating offensive contribution requires balancing scoring output against possession consumption. This is achieved by mapping player performance onto a two-dimensional matrix, plotting Points Per Game (PPG) against True Shooting Percentage (TS%). To evaluate performance in context, the matrix was split into four functional regions using the team's seasonal sample means as reference thresholds: average Points per Game on the horizontal axis and average True Shooting Percentage on the vertical axis. Analysing player output relative to these roster baselines prevents counting statistics from masking possession consumption and clearly distinguishing inefficient volume shooters from true efficient scoring anchors.

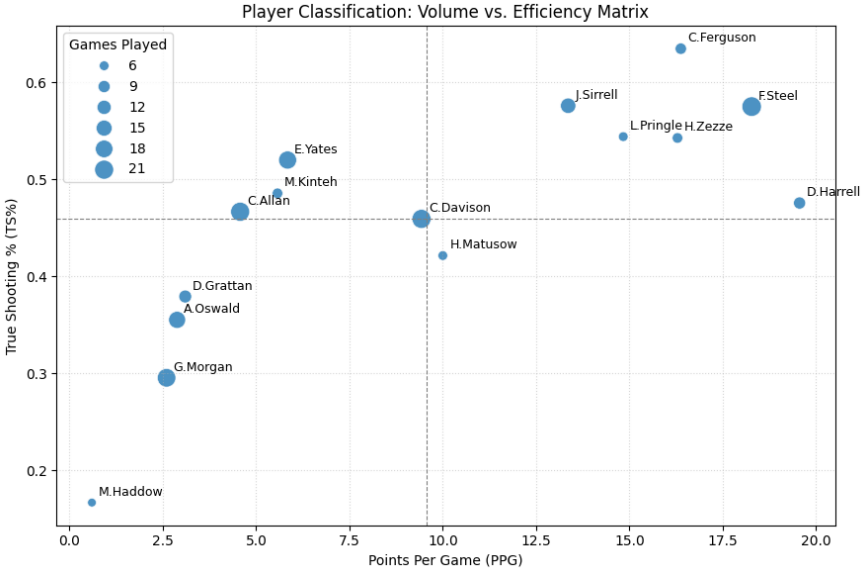



Figure 2: Player Efficiency Matrix: Scoring Volume (PPG) vs. True Shooting Percentage (TS%).

## 3.2 Quadrant Performance Profiles

As illustrated in Figure 2, these threshold lines split the roster into four distinct operational profiles, demonstrating how individual usage impacts overall team scoring quality:

- **Upper-Right: High Volume/High Efficiency (Primary Offensive Anchors):** There are six players who sit comfortably in this quadrant, clearing both the teams average PPG and TS%. While a high density of efficient scorers would typically suggest deep offensive options, contextualising this data against games played (GP) reveals critical roster dynamics. Firstly, in lower game sample sizes, individual TS% is susceptible to small-sample inflation. Brief hot-shooting stretches are less dampened by long-term mean reversion. The presence of six distinct volume/efficiency leaders highlights continuous roster turnover. Instead of one or two established anchors carrying a sustained load across 23 games, the offensive burden was distributed across different players in isolated bursts or specific stretches of the season. On a positive tactical note, it demonstrates a "Next Man Up" attitude in the team. When secondary options were thrust into larger roles due to availability issues, they were capable of producing high-efficiency scoring outputs.

- **Lower-Right: High Volume/Low Efficiency (Inefficient Usage Tax):** This quadrant is notably sparse, populated by only a single player, which remains a highly positive tactical indicator. Offensive possessions are a finite resource, and having minimal high-volume, sub-baseline scorers ensures the team rarely squanders attempts that should be redirected to more efficient options. Furthermore, it reflects strong tactical discipline across the roster, indicating that both the players and coaching staff are successfully managing offensive roles to prevent detrimental shot selection.
- **Upper-Left: Low Volume/High Efficiency (Efficient Role Finishers):** The matrix places three players in this quadrant, driven by distinct skill sets. Two of the three players are interior specialists who generate the majority of their scoring close to the rim. By converting high-percentage shots in the paint, putbacks, and cuts to the basket, these frontcourt players maintain high True Shooting percentages without requiring high usage or perimeter shot creation. The third player operates as a dedicated perimeter specialist. By maintaining strict shot discipline and focusing touch volume on high-value three-point looks, this shooter provides critical spacing for primary handlers while converting low-volume perimeter attempts at an above-average rate. Because these players convert at an elite rate, they show potential for expanded usage. Shifting a few extra possessions to them when primary options face heavy defensive coverage could boost overall team output. The primary caveat is that these players rarely have to create their own shots or shoot under heavy contests, which could impact their efficiency if tasked with an increased volume of unassisted attempts.
- **Lower-Left: Low Volume/Low Efficiency (Limited Usage):** This region represents limited usage profiles, primarily comprising bench rotation, developmental players, and low-minute entries. In small minute samples, single-game shooting variance heavily suppresses a player's seasonal True Shooting percentage, meaning these low metrics often stem from sample volatility rather than systemic inefficiency. Because these players operate at lower scoring volumes, their sub-baseline shooting absorbs very few total team possessions, creating a minimal overall drag on the team's collective True Shooting percentage. From a coaching perspective, these profiles highlight developmental minutes and situational entries. Managing their usage within controlled offensive structures ensures that floor time is utilized for defensive and rotational stability without overexposing them to complex shot-creation responsibilities.

Mapping individual volume against shooting efficiency isolates how different rotational roles contribute to overall scoring production. Ultimately, this framework demonstrates that protecting possession value at the individual level is essential for sustaining a high offensive floor.

# 4 Performance Volatility & Form Tracking

## 4.1 Single-Game Efficiency Volatility

While full-season averages summarize overall production, they obscure short-term consistency and game-to-game performance swings. A player maintaining a 50% season-long True Shooting percentage may achieve this through steady 50% game-by-game outputs or through severe fluctuations between 80% and 20% efficiency. High game-to-game volatility can stem from contextual factors such as early foul trouble, defensive scheme adjustments, or short-term shooting variance. Therefore, judging a player's functional role strictly on isolated single-game outputs can obscure their true contribution. Figure 3 displays the single-game efficiency trajectories for key scorers across

the 2025/26 season. These charts include the two friendlies, so the game numbers count every appearance.

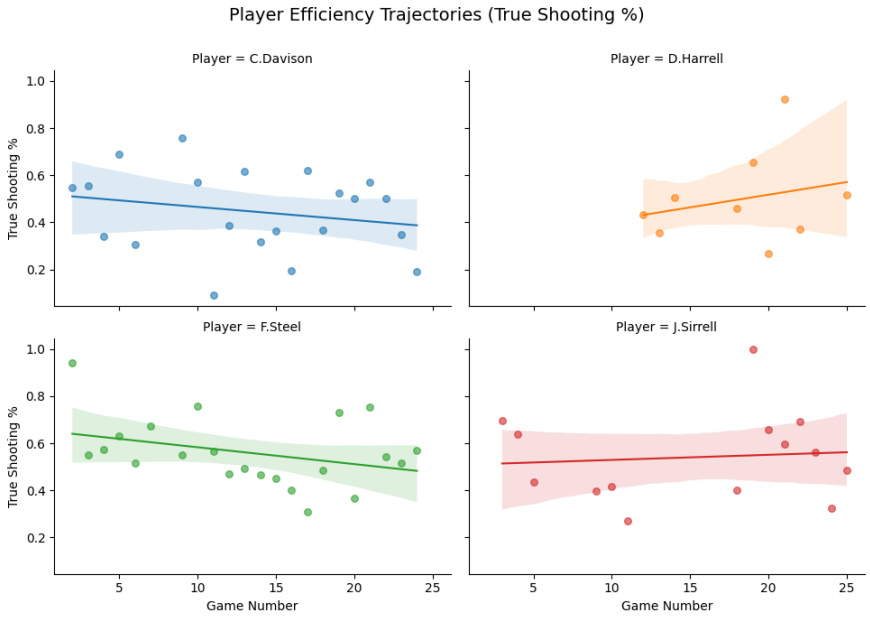



Figure 3: Single-Game True Shooting Percentage (TS%) Trajectories Across the 2025/26 Season.

## 4.2 Moving Average Mechanics ($MA_3$)

To isolate genuine performance trends from routine single-game variance, a backward-looking 3-game moving average ($MA_3$) was applied to the single-game performance metrics of the top four players in total points. Figure 4 displays the relationship between scoring volume and rolling shooting efficiency across sequential games.

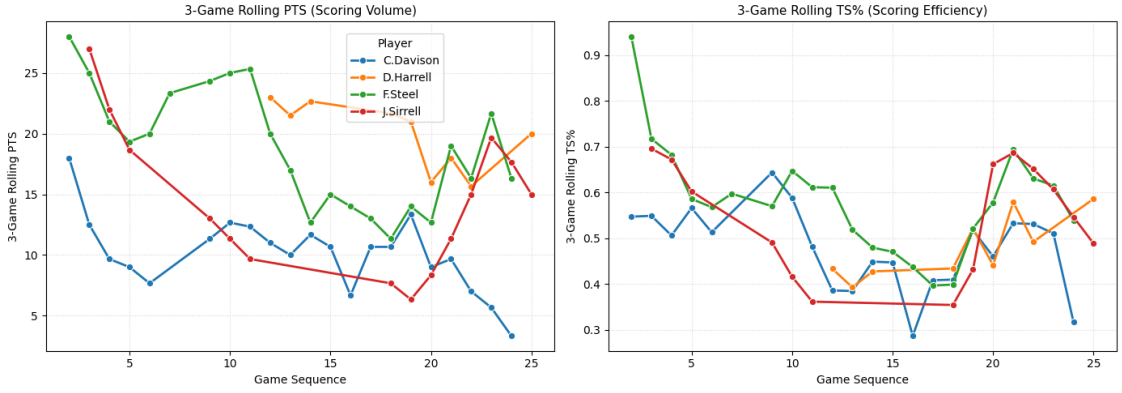



Figure 4: 3-game Rolling Points (PPG) vs. True Shooting Percentage (TS%) Trajectories.

Applying the 3-game window smooths single-game outlier noise while remaining reactive to sustained momentum shifts across consecutive fixtures. The smoothed trends highlight how individual responsibilities adapted to shifting roster availability across the season. Early in the schedule, injury absences among primary options forced the majority of scoring volume onto a single player, resulting in an early-season peak.

As key scorers returned to the line-up, these elevated production levels naturally absorbed a downward correction. Rather than reflecting a drop in individual capability, this decline illustrates a structural rebalancing of offensive usage as primary scorers resumed higher possession shares. Tracking these rolling trajectories confirms that individual output was heavily shaped by tactical context and line-up dynamics rather than isolated individual form.

# 5 Team Performance Drivers & Win Benchmarks

To isolate the macro-level indicators most strongly correlated with victory, team performance metrics were evaluated across all 23 recorded match outcomes for the 2025/26 season (the two friendlies are excluded, as explained in Section 2.1). Analysing victory probabilities across discrete statistical brackets establishes exact operational benchmarks required to secure wins.

## 5.1 Shooting Efficiency Thresholds (TS%)

Shooting efficiency represents a primary determinant of offensive viability. To analyse performance transitions, True Shooting percentages (TS%) were segmented into intervals around a 50% baseline, a solid benchmark distinguishing inefficient from efficient team offence in modern basketball analytics. Figure 5 illustrates team win rates categorised into four True Shooting percentage brackets.

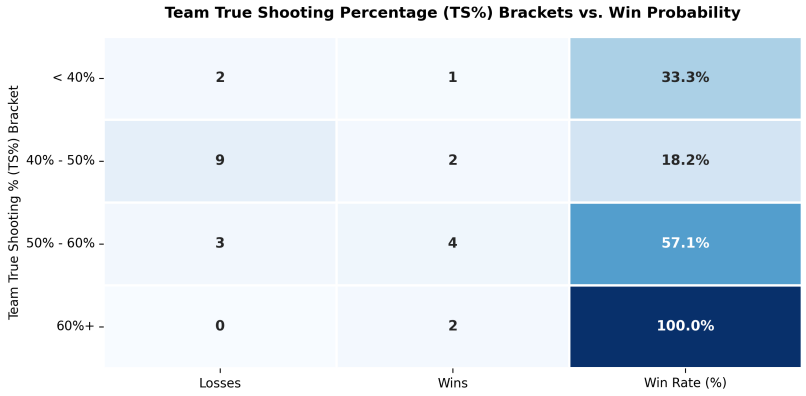


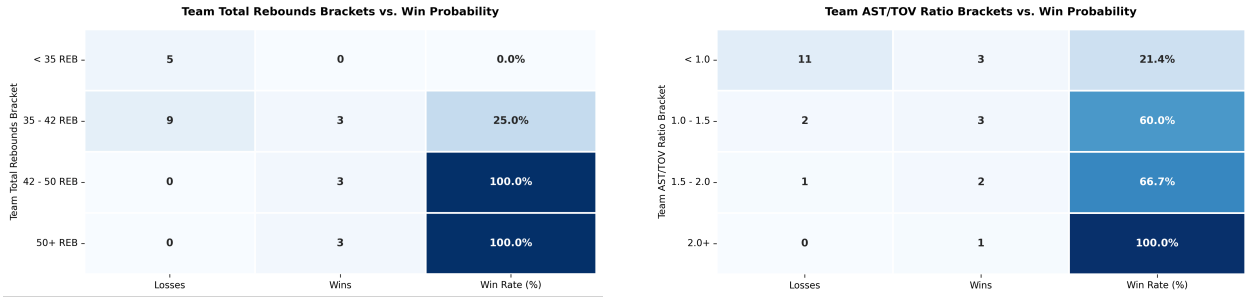

Figure 5: Win Percentage Categorised by Team True Shooting Percentage (TS%).

The dataset demonstrates a critical efficiency tipping point at 50% TS%. When shooting below 40% TS%, the team posted a 33.3% win rate (1 win, 2 losses). Performance compressed further in the 40%-50% efficiency band, dropping to an 18.2% win rate (2 wins, 9 losses). Combined, inefficient shooting games below 50% TS% yielded a low overall win rate of 21.4% (3 wins, 11 losses out of 14 matches).

Conversely, crossing the 50% TS% threshold fundamentally changes match outcomes. Efficiency between 50%-60% pushed the win rate above baseline to 57.1% (4 wins, 3 losses). Above 60% TS%, the team achieved a perfect outcome (2 wins, 0 losses). This steep progression confirms that offensive schemes must prioritise high-value scoring looks, as performances below 50% TS% make securing wins difficult regardless of defensive effort.

## 5.2 Possession Control and Playmaking Stability

Beyond raw shooting execution, winning game outcomes depend heavily on possession balance, specifically through controlling the glass and maintaining ball security. Figure 6 evaluates win probabilities across total rebounding brackets and assist-to-turnover ratio bands.




(a) Rebounding Volume vs. Win Rate

(b) Assist-to-Turnover Ratio vs. Win Rate

Figure 6: Impact of Possession Control and Ball Security on Win Probability.

Rebounding volume acts as a rigid, non-negotiable floor for success. Rebounding targets were binned around a benchmark 42 rebounds, where falling below 35 total rebounds denotes a severely compromised glass presence, and securing 42+ rebounds represents controlling the majority of total game opportunities. When pulled down under 35 total rebounds, the team failed to record a single victory (0.0% win rate, 0 wins, 5 losses). In the intermediate range of 35–42 rebounds, performance remained low at a 25.0% win rate (3 wins, 9 losses). However, reaching 42 or more total rebounds guaranteed victory, achieving a 100.0% win rate across both the 42–50 band (3 wins, 0 losses) and the 50+ band (3 wins, 0 losses). Because second-chance opportunities generally yield high-percentage shots near the basket, limiting the number of offensive rebounds allowed to opponents is pivotal. Securing at least 42 total rebounds effectively starves opponents of second-chance opportunities while providing the possession volume necessary to win.

Playmaking balance yields an equally decisive split. Assist-to-turnover (AST/TOV) brackets were structured around a 1.0 ratio, the fundamental break-even threshold separating possession-negative ball handling from positive offensive creation. In fixtures where the team recorded an AST/TOV ratio below 1.0, the team won just 21.4% of contests (3 wins, 11 losses), suffering from empty possessions and live-ball turnovers. Moving into the 1.0–1.5 band immediately lifted win probability to 60.0% (3 wins, 2 losses), rising to 66.7% (2 wins, 1 loss) between 1.5–2.0 and reaching 100.0% (1 win, 0 losses) when exceeding 2.0. Maintaining an AST/TOV above 1.0 is important to prevent opponent transition points, which directly translate to easy opposition possessions. Limiting uncontested transition opportunities remains a core pillar of playing winning basketball.

## 5.3 Scoring Concentration and Core Production

The distribution of scoring volume across the roster provides a clear benchmark for tactical planning. To analyse top-heavy scoring dependency, combined point totals for the team's top three individual scorers in each fixture were segmented into 10-point brackets around a 55-point baseline. Figure 7 categorises win outcomes based on the combined points generated by the team's top three individual scorers in each game.

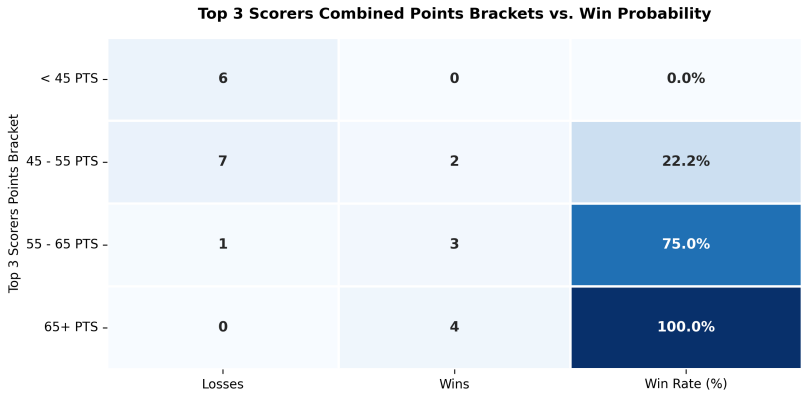



Figure 7: Team Win Percentage Based on Top 3 Player Scoring Concentration.

The data reveals a stark threshold at 55 combined points from the core trio. Scoring under 45 combined points resulted in a 0.0% win rate (0 wins, 6 losses), while scoring between 45-55 points generated a 22.2% win rate (2 wins, 7 losses). Combined point production below 55 points yields a low overall win percentage of 13.3% (2 wins, 13 losses).

Once the top three options meet or break the 55-point mark, victory probability surges. Production in the 55-65 point range produced a 75.0% win rate (3 wins, 1 loss); exceeding 65 points delivered a 100.0% win rate (4 wins, 0 losses). This sharp jump demonstrates that while secondary bench depth is valuable, match outcomes are overwhelmingly governed by whether primary offensive options deliver high-volume scoring outputs. Encouraging the 4th and 5th highest scorers to increase their production slightly may also present an alternative pathway to raise win probability, reducing excessive reliance on core production.

# 6 Predictive Modelling: Decision Tree Classification

While Section 5 evaluated individual performance metrics in isolation, match outcomes in basketball are rarely dictated by a single variable. High scoring volume can be negated by live-ball turnover spikes, just as strong rebounding totals can offset poor perimeter shooting. To capture how these metrics interact dynamically, a machine learning classification model was developed to identify the primary statistical thresholds that govern game outcomes in the 23-game dataset (the two friendlies are excluded, as explained in Section 2.1).

## 6.1 Model Setup & Feature Selection

To construct a predictive framework, key performance indicators were selected as candidate features based on their correlation with match outcomes in Section 5:

- **Top 3 Scorers Points:** Measure of primary offensive volume and scoring concentration.
- **Assist-to-Turnover Ratio:** Measure of playmaking efficiency and ball security.
- **Total Rebounding Volume:** Indicator of glass dominance and possession control.
- **Team True Shooting Percentage (TS%):** Measure of overall shooting efficiency.

A Decision Tree classifier (`DecisionTreeClassifier`) was selected due to its inherent interpretability and non-linear splitting capabilities. The algorithm was allowed to naturally identify the most mathematically decisive performance thresholds to separate wins from losses, producing a clean, actionable decision hierarchy for the coaching staff.

## 6.2 Decision Tree Architecture

Figure 8 displays the trained decision tree architecture, illustrating how recursive binary splitting segments winning games from losing games.

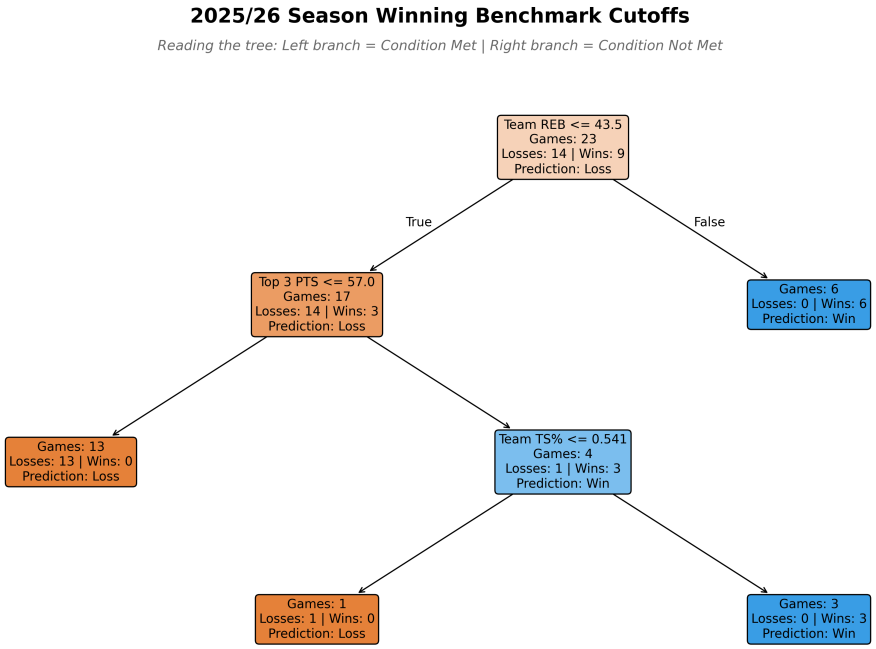



Figure 8: Season Winning Benchmark Decision Flow.

The tree splits first on rebounds: all six games with more than 43.5 were wins, establishing glass control as the absolute gatekeeper for team success. When the team achieves this threshold, they enter a completely undefeated path, posting a clean $6-0$ record (100% Win Rate). Dominating the boards to this degree entirely neutralises other offensive inconsistencies.

Conversely, when rebounding falls below 43.5, match outcomes are dictated by subsequent scoring and shooting thresholds. If the top three scorers fail to combine for more than **57.0** points under these tighter rebounding conditions, the team faces an unrecoverable deficit, resulting in a $0-13$ record. However, when core scoring volume successfully exceeds 57 points despite lower rebounding totals, the game hinges on team shooting efficiency. If shooting efficiency drops below a True Shooting percentage of **54.1**%, the outcome is predicted a loss ($0-1$ record). When efficiency exceeds 54.1% the team successfully converts prolific scoring into victories, securing a $3-0$ record.

## 6.3 Feature Importance & Key Win Determinants

Translating these algorithmic splits into tactical priorities yields three concrete operational benchmarks for game planning:

1. **The Glass-Dominance Floor:** Everything begins with controlling the board. The model demonstrates that securing over 43.5 rebounds creates an automatic winning condition, eliminating the need to rely heavily on tight offensive margins.
2. **Core Scoring and Efficiency Pairing:** When rebounding volume sits below the elite 43.5 threshold, victory becomes entirely dependent on offensive execution. The team must ensure primary options combine for at least 58 points while maintaining team shooting efficiency above 54.1%.
3. **Eliminating Low-Production Vulnerability:** Failing to clear core scoring thresholds while simultaneously losing the battle of the boards creates an unrecoverable structural deficit (0–13 record), proving that offensive stagnation cannot be overcome without defensive rebounding support.

The three benchmarks selected for the analysis were Team_REB, Top3_PTS, and Team_TS%, as these were identified by the decision tree as the features contributing most strongly to the prediction of player efficiency. Team_REB had the greatest feature importance (54.9%), followed by Top3_PTS (31.4%) and Team_TS% (13.7%), together accounting for 100% of the model's importance. In contrast, Team_AST_TOV had an importance of 0%, indicating that it did not contribute to any of the decision splits used by the model. Therefore, the three selected benchmarks provide the most meaningful basis for further analysis of the factors associated with player efficiency.

# 7 Strategic Coaching Recommendations & Conclusion

This analysis demonstrates that match outcomes across the 2025/26 season were governed by distinct, measurable operational thresholds rather than random variance. By bridging the gap between raw statistical output and tactical application, these findings translate quantitative models into an evidence-based roadmap for game preparation and roster management.

## Operational Checklist for Game Day

Tactical preparation must centre on three primary quantitative objectives. The absolute priority is controlling the glass; securing more than 43.5 rebounds creates an automatic winning environment by entirely insulating the team from offensive slumps and eliminating opponent second-chance opportunities. When dominating the boards to this degree is not achievable, the focus must immediately shift to core scoring. The offensive scheme must ensure the top three options receive the required shot volume and spacing to combine for at least 57 points. Finally, on nights when rebounding margins are tight and primary scorers get enough shots, the squad must focus on rigid execution discipline. This requires maintaining a True Shooting percentage above 54.1%.

## Rotational Adjustments

To consistently hit these game-day benchmarks, minute distribution must reflect possession value. The player profiling matrix highlights a heavy reliance on primary scoring anchors, but it also identifies highly efficient, low-volume finishers operating both on the perimeter and in the paint. Shifting a controlled increase in touches to these secondary options when the main trio faces heavy defensive coverage can sustain the team's offensive floor without sacrificing efficiency. Furthermore, developmental players and low-minute entries must be managed within strict offensive structures. Limiting their complex shot-creation responsibilities ensures their floor time provides necessary

rotational rest for the primary scorers, without exposing the team to live-ball turnovers and subsequent defensive vulnerabilities.

### Project Scope Limitations

While these quantitative thresholds provide a robust baseline for tactical planning, the predictive model is inherently constrained by the 23-game sample size. The current dataset successfully isolates internal team performance metrics, but it does not explicitly quantify opponent strength, strength of schedule, or specific defensive matchups. The model currently operates under the assumption of a baseline level of opponent resistance across all fixtures. Future iterations of this analytical framework would benefit from integrating opponent tracking data and defensive efficiency ratings, allowing for a more comprehensive understanding of how specific opposition schemes impact the team's established offensive benchmarks.